DFIG Wind Energy Conversion System

Doubly Fed Induction Generator Trainer — Technical Documentation

ENGINEERING COLLEGE LABORATORY EQUIPMENT MANUAL

Rotor Side Converter with STM32F407VGT6-based DQ Current Control
Document Version 1.0

Table of Contents

1. Introduction

2. System Overview

3. Hardware Specifications

4. Working Principle of the DFIG

5. Rotor Side Converter (RSC) & Control Algorithm

6. Protection and Safety Features

7. STM32F407VGT6 Controller Board

8. Conclusion

9. Appendix A — Consolidated Specification Tables

10. Appendix B — Glossary of Terms

1. Introduction

1.1 Purpose of this Document

This document describes the DFIG (Doubly Fed Induction Generator) Wind Energy Conversion System trainer installed as laboratory equipment. It is intended to give students and laboratory instructors a clear understanding of the equipment's construction, its working principle, and the control strategy used to operate it, so that the trainer can be used effectively for hands-on learning in electrical machines, power electronics, and renewable energy courses.

1.2 What is a DFIG-based Wind Energy Conversion System

A Doubly Fed Induction Generator (DFIG) system is one of the most widely used configurations in modern variable-speed wind turbines. Unlike a conventional induction generator, whose stator alone is connected to the grid, a DFIG has both its stator and its rotor connected to external circuits: the stator is connected directly to the grid, while the rotor (through slip rings) is connected to a power electronic converter. This arrangement allows the machine to operate efficiently over a wide range of wind speeds — both below and above its natural synchronous speed — while only a fraction of the total power (typically 25–30%) needs to flow through the power electronic converter. This makes DFIG systems more cost-effective than fully-converter-fed alternatives for large-scale wind turbines.

In a real wind turbine, the rotor blades and gearbox drive the induction generator at variable speed depending on wind conditions. In this laboratory trainer, that variable-speed prime mover is emulated using a separately excited DC motor, whose speed can be controlled to mimic different wind conditions, while the actual power conversion and control behaviour is demonstrated using a genuine slip-ring induction machine and a rotor-side power converter.

1.3 Learning Objectives

Through this trainer, students are expected to gain practical understanding of:

- The construction and operating principle of a slip-ring (wound rotor) induction machine used as a DFIG.
- Sub-synchronous and super-synchronous modes of operation.
- The role and internal structure of a Rotor Side Converter (RSC).
- DQ (Park's transformation) based current control of a rotor-side converter.
- Protection schemes used in IGBT-based power converters.
- Use of an ARM Cortex-M4 (STM32F407) microcontroller for real-time motor/generator control.

2. System Overview

The trainer consists of three major sub-systems working together, as shown in Figure 1:

1. **Prime Mover Emulator:** A 2 HP, 220 V DC shunt/separately-excited motor with double-side shaft extension, used to drive the induction machine at variable speed, emulating the variable-speed input a wind turbine rotor would provide.
2. **DFIG (Slip Ring Induction Machine):** A 1 HP, three-phase, 415 V slip-ring induction motor, mechanically coupled to the DC motor, and electrically connected so that its stator supplies power directly (analogous to the grid connection) while its rotor is connected to the Rotor Side Converter.
3. **Rotor Side Converter (RSC) and Controller:** A two-level IGBT-based inverter connected to the rotor circuit through the slip rings, controlled by an STM32F407VGT6 microcontroller running a DQ-based current control algorithm.

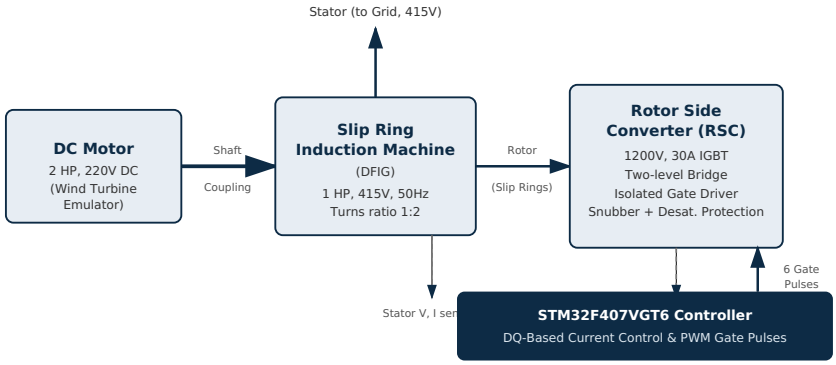


Figure 1: Block diagram of the DFIG Wind Energy Conversion System trainer

2.1 Mechanical Coupling

The double-side shaft extension of the DC motor allows a coupling on one side to the slip-ring induction machine and, optionally, instrumentation such as a tachometer or torque transducer on the other side, enabling speed and torque measurement during experiments.

2.2 Electrical Configuration

The stator of the induction machine is wound for 415 V three-phase operation and is rated for 1.4 A per phase. The rotor is wound with a stator-to-rotor turns ratio of 1:2, which means rotor-side voltages are transformed relative to the stator according to this ratio and the operating slip — this is an important parameter when sizing and interpreting the rotor-side converter's voltage and current ratings.

3. Hardware Specifications

3.1 DC Motor (Prime Mover / Wind Turbine Emulator)

Parameter	Specification
Type	Separately excited / shunt DC motor
Power rating	2 HP
Armature voltage	220 V DC
Field voltage	220 V DC
Rated speed	1750 RPM
Shaft configuration	Double side shaft extension

3.2 Slip Ring Induction Motor (Operating as DFIG)

Parameter	Specification
Type	Three-phase wound-rotor (slip ring) induction machine
Power rating	1 HP
Frequency	50 Hz
Stator voltage / current	Three-phase, 415 V / 1.4 A
Stator-to-rotor turns ratio	1 : 2
Rated speed	1410 RPM

3.3 Rotor Side Converter (RSC) Hardware

Parameter	Specification
Converter topology	Two-level voltage source inverter, rotor-circuit connected
Driver circuit	Infineon IGBT gate driver
Power module	1200 V, 30 A three-phase IGBT module with heat sink
Protection circuit	Snubber circuit across IGBT module
Gate drive isolation	Isolated driver circuit with in-built power supply, 6 gate pulse outputs
Overload/short-circuit protection	IGBT desaturation ($V_{CE(sat)}$) protection
Status indication	READY and FAULT LED indication
Fault handling	Latching of driver output on fault, manual RESET button to clear
Sensing	Three rotor currents, stator voltages, stator currents

3.4 Controller Board — STM32F407VGT6

Parameter	Specification
Core	ARM Cortex-M4F, 32-bit
Clock speed	168 MHz
Flash memory	1 MB

RAM	192 KB
Package	LQFP100
Programming/debug	On-board ST-LINK/V2 with mode-selection switch
Power supply	USB bus-powered or external supply
Peripherals	On-board ADC, DAC, USB and JTAG terminals

4. Working Principle of the DFIG

4.1 Variable-Speed Operation

In a DFIG, the mechanical speed of the rotor does not need to match the synchronous speed set by the grid frequency and the number of poles. Instead, the rotor-side converter injects a rotor current at slip frequency, allowing the machine to operate at speeds above or below synchronous speed while the stator frequency remains fixed to the grid frequency.

- **Sub-synchronous operation** (rotor speed < synchronous speed): power flows from the rotor circuit into the rotor-side converter, and the machine absorbs rotor power from the converter.
- **Super-synchronous operation** (rotor speed > synchronous speed): power flows from the rotor to the converter and is fed back, typically toward the grid side, in a full bidirectional back-to-back converter system.

In this trainer, this variable-speed behaviour is created by driving the induction machine's shaft with the DC motor at different speeds around the induction machine's synchronous speed, while the rotor-side converter, under STM32-based DQ control, regulates the rotor currents accordingly.



Figure 2: Power flow paths in sub-synchronous and super-synchronous DFIG operation

4.2 Role of the Turns Ratio

Since the stator-to-rotor turns ratio is 1:2, the induced rotor EMF at standstill would be twice the stator EMF per phase referred through the winding ratio; at any operating slip s, the actual rotor voltage is proportional to s times this ratio. This relationship is used when selecting the rotor-side converter's voltage rating and when interpreting current/voltage sensor readings during experiments.

5. Rotor Side Converter (RSC) & Control Algorithm

5.1 Purpose of the RSC

The Rotor Side Converter is connected between the rotor windings (via slip rings) and a DC bus, and its role is to control the magnitude, frequency, and phase of the rotor currents. By controlling these rotor currents, the RSC indirectly controls the active and reactive power delivered by the stator, as well as the machine's rotor speed / torque behaviour.

5.2 DQ-Based Current Control Scheme

The control strategy used is a DQ (direct-quadrature) current control scheme, based on Park's transformation. The three-phase rotor currents (and relevant stator quantities) are transformed from the stationary three-phase reference frame into a rotating two-axis (d-q) reference frame, typically aligned with the stator flux or stator voltage vector. In this rotating frame:

- The **d-axis current component** is generally used to control reactive power / magnetizing behaviour.
- The **q-axis current component** is generally used to control active power / electromagnetic torque.

Each axis current is regulated using a closed-loop current controller (typically PI-based), and the resulting reference voltages are transformed back into the three-phase stationary frame to generate switching signals for the IGBT bridge via Pulse Width Modulation (PWM).

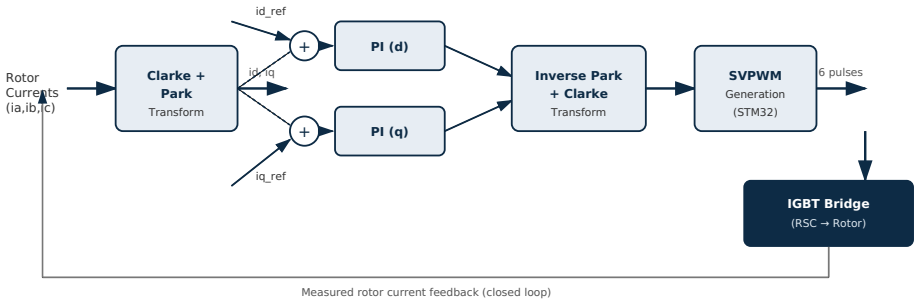


Figure 3: DQ-based rotor current control loop implemented on the STM32F407VGT6

5.3 Sensing Requirements

To implement this control scheme, the system continuously senses:

- Three rotor currents (for the inner current control loop, transformed into d-q components).
- Stator voltages (used for flux/voltage-oriented reference frame alignment and synchronization).
- Stator currents (used for monitoring and, where implemented, outer power control loops).

5.4 Gate Pulse Generation

The STM32F407VGT6 generates the six gate pulses required to drive the two-level, three-phase IGBT bridge (two switches per leg, three legs). These pulses are sent through the isolated gate driver circuit, which provides the necessary isolation and drive power for the IGBTs while also monitoring for desaturation faults.

6. Protection and Safety Features

6.1 Desaturation Protection

Desaturation protection monitors the collector-emitter voltage of each IGBT during conduction. If a device is subjected to an overload or a short-circuit condition, it comes out of full saturation and its V_{CE} rises; the driver circuit detects this abnormal rise and rapidly turns off the gate signal to prevent device damage.

6.2 READY / FAULT Indication and Latching

The system provides visual indication of its operating state:

- **READY** indication shows the converter is powered and healthy, with no active fault.
- **FAULT** indication shows that a protection trip (e.g., desaturation, overcurrent) has occurred.

Once a fault occurs, the driver output is latched OFF to prevent the converter from re-energizing automatically. The operator must press the **RESET** button to clear the latched fault and restore normal operation, ensuring a fault condition is deliberately acknowledged before re-starting the converter — an important safety practice in a teaching lab environment.

6.3 Snubber Circuit

The snubber circuit connected across the IGBT module limits voltage spikes (dv/dt) during switching transitions, protecting the power devices from transient overvoltage stress caused by stray inductance in the power circuit.

> **Laboratory Safety Note:** Students and instructors should always verify the READY indication before energizing the converter, and should never bypass or ignore a FAULT/latch condition without first identifying the underlying cause.

7. STM32F407VGT6 Controller Board

The control algorithm for the RSC is implemented on an STM32F407VGT6 development board, built around the ARM Cortex-M4F core. This controller family is widely used in motor control and power electronics applications because it combines a floating-point unit (FPU), high clock speed, and rich set of on-chip peripherals suited to real-time control.

7.1 Key Features Relevant to This Application

Feature	Relevance to RSC Control
168 MHz Cortex-M4F core	Executes DQ transformations, PI control loops, and PWM updates within the required control-loop timing
On-board ADC	Samples rotor currents, stator voltages, and stator currents
On-board DAC	Can output analog monitoring signals for oscilloscope observation of internal control variables
1 MB Flash / 192 KB RAM	Stores control firmware and real-time data buffers
ST-LINK/V2 on-board	Allows direct programming and debugging without an external programmer
USB / JTAG terminal	Supports firmware upload, live debugging, and monitoring

7.2 Programming and Debugging

The on-board ST-LINK/V2 interface, along with a mode-selection switch, allows the board to be used either in a standalone programming mode or a debug mode where breakpoints and live variable inspection are possible — a useful feature for students learning to tune current controllers or observe transformation stages during experiments.

8. Conclusion

This DFIG Wind Energy Conversion System trainer provides a compact but functionally complete platform for studying variable-speed wind generation technology. By combining a DC motor emulated prime mover, a genuine slip-ring induction machine, and an IGBT-based rotor-side converter controlled through DQ-based current control on an STM32F407VGT6 microcontroller, the system allows students to observe, measure, and control the same fundamental behaviours found in commercial DFIG-based wind turbines — from sub/super-synchronous operation to real-time power converter control and protection.

Proper understanding of the specifications, working principle, and protection features documented here is recommended before students operate the equipment, in order to ensure safe and effective use of the trainer.

Appendix A — Consolidated Specification Tables

A.1 DC Motor

Power	Armature Voltage	Field Voltage	Speed	Shaft
2 HP	220 V DC	220 V DC	1750 RPM	Double side extension

A.2 Slip Ring Induction Motor (DFIG)

Power	Frequency	Stator Voltage/Current	Turns Ratio (S:R)	Speed
1 HP	50 Hz	415 V, 1.4 A (3-phase)	1:2	1410 RPM

A.3 Rotor Side Converter

Topology	Module Rating	Driver	Protection	Sensing
2-level, 3-phase VSI	1200 V, 30 A IGBT	Infineon, isolated, 6 gate pulses	Desaturation, snubber, latch+reset	3 rotor currents, stator V & I

A.4 Controller Board

MCU	Clock	Flash	RAM	Package	Debug
STM32F407VGT6	168 MHz	1 MB	192 KB	LQFP100	On-board ST-LINK/V2

Appendix B — Glossary of Terms

DFIG
Doubly Fed Induction Generator — an induction machine with both stator and rotor electrically accessible, used in variable-speed wind turbines.

RSC
Rotor Side Converter — the power electronic converter connected to the rotor circuit of the DFIG through slip rings.

IGBT
Insulated Gate Bipolar Transistor — a power semiconductor switch used in the converter bridge.

DQ Transformation
Park's transformation, converting three-phase AC quantities into two DC-like quantities (d-axis and q-axis) in a rotating reference frame, simplifying control design.

Slip
The normalized difference between synchronous speed and actual rotor speed of an induction machine.

Sub-synchronous operation
Operation where the rotor speed is below the synchronous speed corresponding to the grid frequency.

Super-synchronous operation
Operation where the rotor speed is above the synchronous speed corresponding to the grid frequency.

Desaturation Protection
A protection method that detects abnormal rise in IGBT collector-emitter voltage during conduction, indicating overload or short-circuit, and trips the gate drive.

Snubber Circuit
An auxiliary RC or RCD circuit connected across a power switching device to limit voltage transients during switching.

PWM
Pulse Width Modulation — the switching technique used to synthesize the desired AC output voltage from a DC bus using power semiconductor switches.

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